

(4.2)

$$p(x) = a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0, a_i \in \mathbb{R} \forall i \in \{0, \dots, 4\}$$

$$p(0) = a_4 \cdot 0 + a_3 \cdot 0 + a_2 \cdot 0 + a_1 \cdot 0 + a_0 = 1$$

$$\underline{a_0 = 1}$$

$$p(1) = a_4 \cdot 1 + a_3 \cdot 1 + a_2 \cdot 1 + a_1 \cdot 1 + a_0 = 2$$

$$a_4 + a_3 + a_2 + a_1 + 1 = 2$$

$$\underline{a_4 + a_3 + a_2 + a_1 = 1}$$

$$p(2) = a_4 \cdot 16 + a_3 \cdot 8 + a_2 \cdot 4 + a_1 \cdot 2 + a_0 = -3$$

$$16a_4 + 8a_3 + 4a_2 + 2a_1 + 1 = -3$$

$$16a_4 + 8a_3 + 4a_2 + 2a_1 = -4 \quad | :2$$

$$\underline{8a_4 + 4a_3 + 2a_2 + a_1 = -2}$$

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \end{pmatrix}, X = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}, b = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

Použijeme řešení s nejmenší normou.

$$x_{\text{lm}} = A^T (A A^T)^{-1} b$$

$$A A^T = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 4 & 8 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \\ 1 & 8 \end{pmatrix} = \begin{pmatrix} 4 & 15 \\ 15 & 85 \end{pmatrix}$$

$$(A A^T)^{-1} = \frac{1}{115} \begin{pmatrix} 85 & -15 \\ -15 & 4 \end{pmatrix}$$

$$(A A^T)^{-1} b = \begin{pmatrix} 85 & -15 \\ -15 & 4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{23}{115} \end{pmatrix}$$

$$x_{\text{lm}} = A^T \cdot \begin{pmatrix} 1 \\ -\frac{23}{115} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \\ 1 & 8 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -\frac{23}{115} \end{pmatrix}$$

$$\Rightarrow a_1 = \frac{92}{115}, a_2 = \frac{69}{115}, a_3 = \frac{23}{115}, a_4 = -\frac{69}{115}$$

Výsledkem je polynom:

$$p(x) = -\frac{69}{115} x^4 + \frac{23}{115} x^3 + \frac{69}{115} x^2 + \frac{92}{115} x + 1$$